

Generalized User Representations for Large-Scale Recommendations and Downstream Tasks

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Abstract

Accurately capturing diverse user preferences at scale is a core challenge for large-scale recommender systems like Spotify's, given the complexity and variability of user behavior. To address this, we propose a two-stage framework that combines representation learning and transfer learning to produce generalized user embeddings. In the first stage, an autoencoder compresses rich user features into a compact latent space. In the second, task-specific models consume these embeddings via transfer learning, removing the need for manual feature engineering. This approach enhances flexibility by allowing dynamic updates to input features, enabling near-real-time responsiveness. The framework has been deployed in production at Spotify with an efficient infrastructure that allows downstream models to operate independently. Extensive online experiments in a live setting show significant improvements in metrics such as consumption share, content discovery, and search success. Additionally, our method achieves these gains while substantially reducing infrastructure costs.

Keywords

user model, embeddings, recommender systems, cold-start models

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1 Introduction

Online music streaming services have grown increasingly popular in recent years. Music recommender systems face different challenges than traditional recommender systems. The tracks are often short, and several tracks can be played in a single session without any feedback [11]. Users also have conflicting desires: revisiting favorite tracks while discovering new music to diversify their experiences [10, 15]. Furthermore, relying on interaction and consumption as primary signals for training recommender systems makes it particularly challenging to provide recommendations for new users [12, 13]. Despite recent advancements in user modeling, accurately capturing and representing user interests in large-scale music streaming services remains a challenging task [9]. Treating each task independently (e.g., retrieval from long-term taste, bandits for new users) can increase system complexity and lead to disjointed early user experiences due to uncoordinated model interactions [2, 7].

In this paper, we detail how our model builds an effective user representation, address the challenges associated with this framework, and demonstrate how it captures both holistic user interests and current preferences. Our contributions aim to answer the following three questions:

- [RQ1] How can we efficiently design a model to capture rich user representation in a large-scale recommender system that encompasses core user interests while being adaptable to various downstream tasks?

- [RQ2] How can we design a user representation model that works for cold-start users and improves as users become more established on the platform? and
- [RQ3] How can we devise effective evaluation strategies to measure the efficacy of the vector embedding space to ensure value across diverse downstream tasks?

We tackle these questions using a two-stage approach. In the first stage, we develop an autoencoder model that ingests a range of user features, such as listening history, demographics, and contextual signals, to learn rich, generalized user representations. In the second stage, these representations are leveraged in a real-world transfer learning setting, where they are adapted to support a variety of downstream applications. We show that our method offers significant advantages over standard baselines. By enabling a shared and adaptable user embedding, our approach supports diverse tasks across the platform. Using internal Spotify data, we demonstrate strong performance and provide an analysis of each component’s contribution. This system is now deployed at scale in production.

2 Approach

Figure 1 presents an overview of the architecture of our generalized user representation model. Initially, modality encoders preprocess the track features and represent each track with 80-dimensional vectors derived from co-occurrence statistics and audio features. This embedding space has been shown to perform well for music recommendation tasks [1, 4]. These track embeddings, combined with other user data such as demographics, onboarding signals, and contextual information, form the input features used in the main model training to generate the user representations. The resulting learned representations can then be used as feature inputs for a variety of recommendation tasks. This preprocessing steps enables faster training, higher efficiency and low-latency for training and serving the resulting representations.

3 Transfer Learning

Our approach supports large-scale recommendation tasks such as retrieval, ranking, and generation, using a generalized user representation that transfers well across systems.

Responsiveness: Fast inference alone does not guarantee responsiveness to changes in user preferences, especially when relying on stale batch features. To address this, we employ Near-Real-Time (NRT) inference triggered by user activity events. These events are pre-processed, passed through the model, and stored in an on-line feature store within minutes. By combining NRT with batch inference, we support both active and inactive users effectively.

Cold-Start Awareness: During onboarding, users may select preferred artists or languages. These signals are encoded and passed through the same model used for existing users, enabling immediate personalization. We combine onboarding features with demographic information and gradually transition to behavior-based inputs as more data becomes available.

Stability: For effective transfer learning, user representations must evolve within stable vector spaces that preserve semantic meaning over time. Although periodic retraining is necessary to

Table 1: Performance for predicting listening within next seven days for established users using generalized user representation.

Comparison Vs	Accuracy	AUC
NMF	+15.2%	+18.6%
PMF	+10.1%	+12.7%
LightFM	+2.3%	+3.9%
DLRM	+1.5%	+2.8%
VAE-CF	+4.0%	+5.7%
average embeddings	+1.8%	+1.6%

combat model drift [8], unsynchronized updates can disrupt downstream dependencies. To mitigate this, we implement Batch Management, a coordinated retraining strategy that assigns a unique batch ID to each update. Downstream models are retrained in sync, ensuring that embeddings are only compared within the same batch. Both training and inference operate within a consistent batch context, while production systems continue running on a “legacy” batch during updates. This approach guarantees synchronization, consistent comparisons, and uninterrupted service, preserving stability across the entire model pipeline.

4 Empirical Studies

We evaluate our framework across a range of tasks and baselines, targeting both cold-start and established users. The training dataset is constructed using audio streaming data drawn from Spotify global user base. For evaluation, we sample over five million users at random, ensuring a representative distribution by considering attributes such as country of registration, subscription status (free or paid), and account age. Online performance is assessed through A/B tests conducted on production traffic.

To measure the effectiveness of our generalized user representations, we first examine their performance in future track prediction tasks. For established users, we use a 7-day prediction window and compare our approach to several baseline models, including matrix factorization techniques (NMF [14], PMF [5]), deep learning models (LightFM [2], DLRM [6]), VAE-CF [3], and embeddings averaged from users’ listening histories. All models are aligned in terms of embedding dimensionality. As reported in Table 1, our model consistently outperforms these baselines, validating its predictive power and supporting our first research question (RQ1). Notably, our results show improvements even over deep learning-based methods such as LightFM and DLRM, as well as autoencoder-based approaches like VAE-CF. Interestingly, average embeddings perform reasonably well compared to other baselines, which we attribute to the influence of popularity bias that is a dominant factor in aggregated listening behavior.

For cold-start scenarios, we evaluate performance over a 4-hour window. Here, our method is compared to baselines that average onboarding artist embeddings or use popular tracks inferred from demographic features. Results in Table 2 show that our unified representation provides clear improvements, effectively capturing early user intent even in the absence of complete onboarding information. This supports RQ2 and demonstrates the model robustness in sparse data settings.

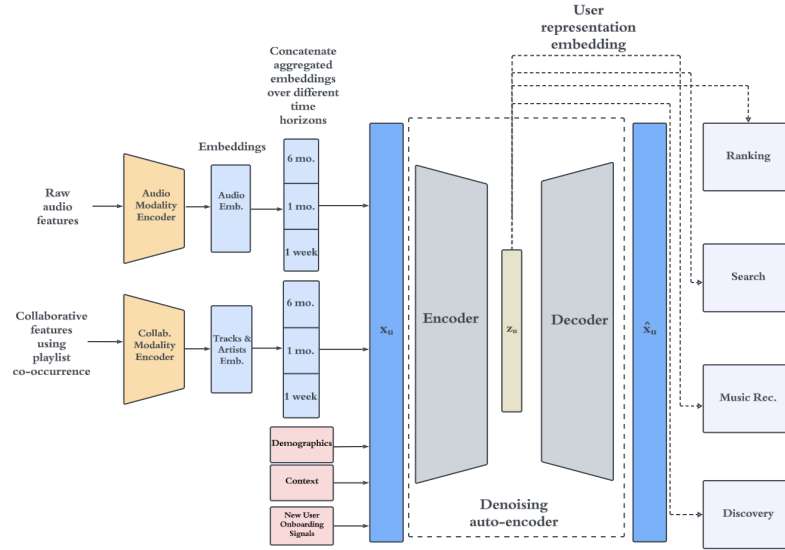


Figure 1: Generalized User Representation Model Architecture. Catalog interactions are embedded using pre-trained modality encoders, which then serve as input to the autoencoder model. The resulting representation serves as the User Representation for various downstream recommendation tasks.

Table 2: Performance for predicting listening within four hours following user registration.

Comparison Vs	Onboarding Status	Accuracy	AUC
popularity heuristic	Completed	+26.2%	+27.0%
popularity heuristic	Not Completed	+24.6%	+24.7%
average embeddings	Completed	+5.0%	+5.1%

Table 3: Performance at finding similar clusters for user representation vs. average embeddings.

Cluster Heuristic	nDCG@50
Same favorite artists	+2.9%
Same country of most listened artists	+5.5%
Same new user onboarding	+26.2%

We also conduct a clustering-based evaluation to assess the intrinsic quality of our embeddings. By comparing nDCG@50 values obtained through nearest-neighbor lookups, we find that our representation outperforms those derived from averaged item embeddings across diverse user clusters. These findings, presented in Table 3, lend support to RQ3 and highlight the model ability to group users in behaviorally meaningful ways.

Finally, we test the applicability of our user representations in real-world scenarios by integrating them into downstream production models. Our offline experiments and online A/B tests demonstrate that the learned embeddings provide meaningful transferability and measurable impact when deployed in live systems. Table 4 provides a summary of the key results achieved across various applications of the generalized user model.

Candidate generation. The model powers the generation of album and playlist candidates in Home page shelves. It led to significant increases in album discoveries and impression-to-stream (i2s) rates, reinforcing the value of using a generalized user representation for early-stage retrieval.

Search. Used to re-rank search results to personalize the search results, the model yielded a +0.06% overall improvement and a +0.76% increase in podcast search success, which highlights effectiveness across modalities. These gains, particularly in a highly optimized system, underscore the model ability to effectively capture cross-modal user behaviors through its embeddings.

Home Ranking Model. Applied to rank item shelves on the Home page, the model enhanced music discovery and increased the share of Home content consumption. The results show a successful shift in user engagement from familiar content toward new discoveries, reflecting the model ability to encourage broader exploration.

Artist preference modeling. This captures user-artist affinity and is leveraged across several recommendation and ranking scenarios. We achieved a 50% reduction in infrastructure and feature-related costs without affecting top-line metrics, resulting in simplified models, reduced technical debt, and faster deployment cycles.

To investigate the relative importance of features, we performed an ablation study on the input features. We removed specific features to assess their impact. When we exclude new user onboarding signals, we observe a 13.8% decrease in nDCG@50 for clusters formed using the same new user onboarding data. This indicates that without onboarding signals, the model’s ability to identify new

Table 4: Downstream performance using generalized user embeddings: Online results from candidate generation and search models, as well as end-to-end performance in the Home Ranking model, demonstrate the effectiveness of transfer learning.

Downstream Model	Metric 1	Metric 2
Candidate Generation Model	Discoveries +2.9%	Shelf level i2s +13%
Search Model	Overall Search Success +0.06%	Podcast Success +0.76%
Home Ranking Model	Music Discovery Success +0.20%	Consumption Share +.05%

user clusters is diminished. Excluding modality encoder embeddings results in a 4.2% decrease in AUC for the future listening evaluation over 7 days. Additionally, we see a 37.1% drop in nDCG@50 for clusters formed based on favorite artists. This highlights the importance of modality encoder embeddings to the system success. Finally, omitting user based static features, such as the country of registration, leads to a 12.1% decrease in nDCG@50 for clusters formed based on the country of most listened artists.

5 Conclusion

We presented a scalable framework for learning generalized user representations, now deployed in production at Spotify to support tasks like personalized search, candidate generation, and ranking. Combining autoencoder-based representation learning with transfer learning, our approach performs strongly across both cold-start and established user scenarios, while reducing infrastructure complexity. Future work includes integrating additional modalities—such as lyrics, playlists, and metadata—and leveraging large language model (LLM) embeddings to enhance semantic understanding. We also plan to extend this framework to domains like podcasts, news, and commerce for broader personalization.

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